

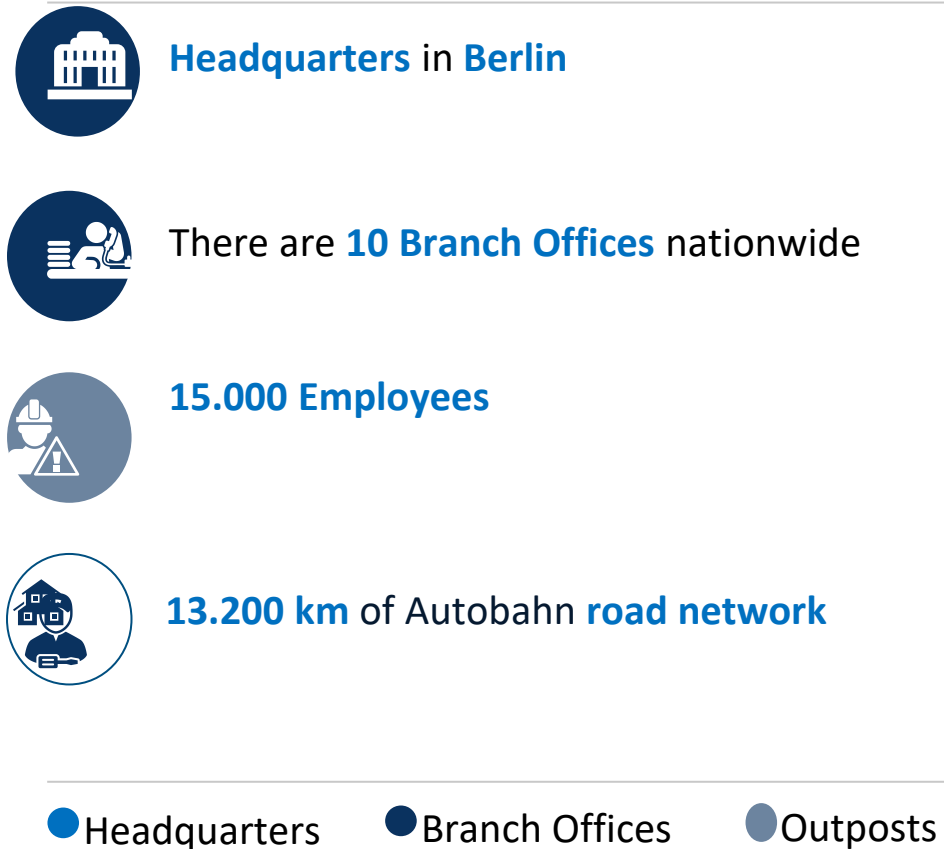
Automated Traffic Forecasting for Alpine Holiday Corridors A 8 East & A 93 South

TUM Science Hackathon 2026

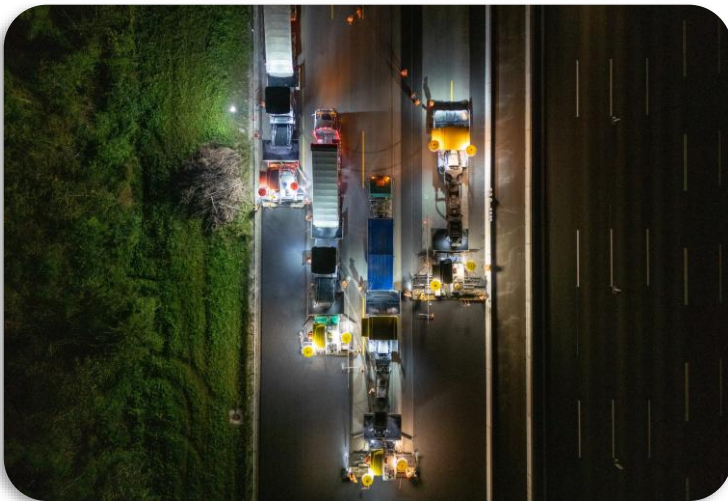
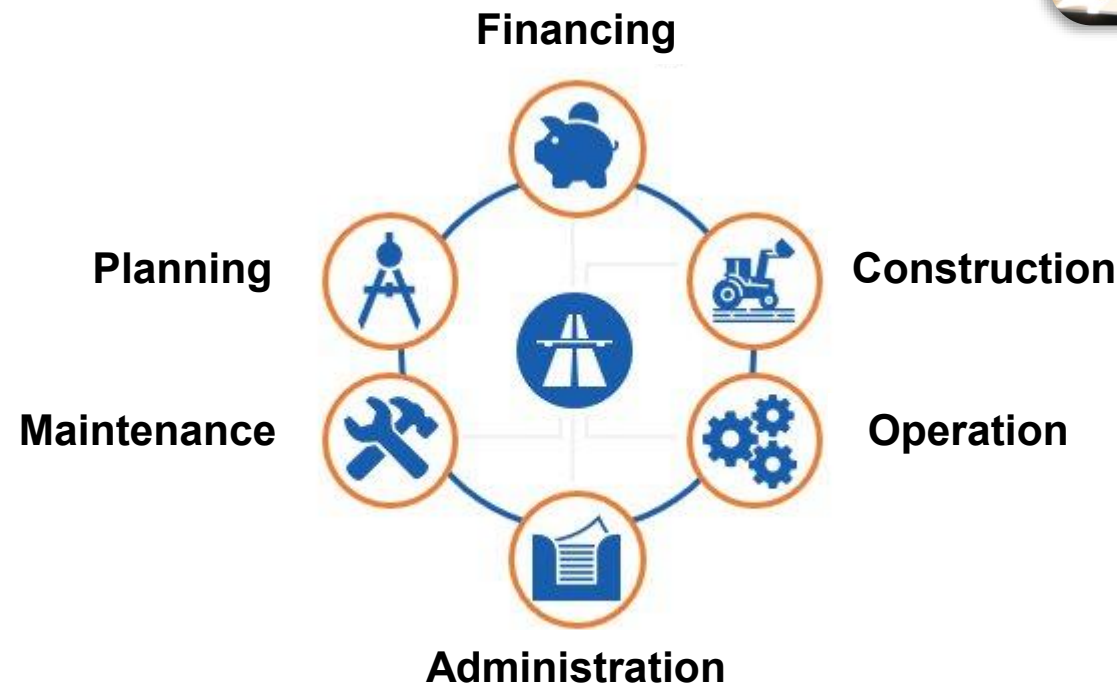
Quelle: Autobahn GmbH | Felix Bonn | A8 Ost

www.autobahn.de

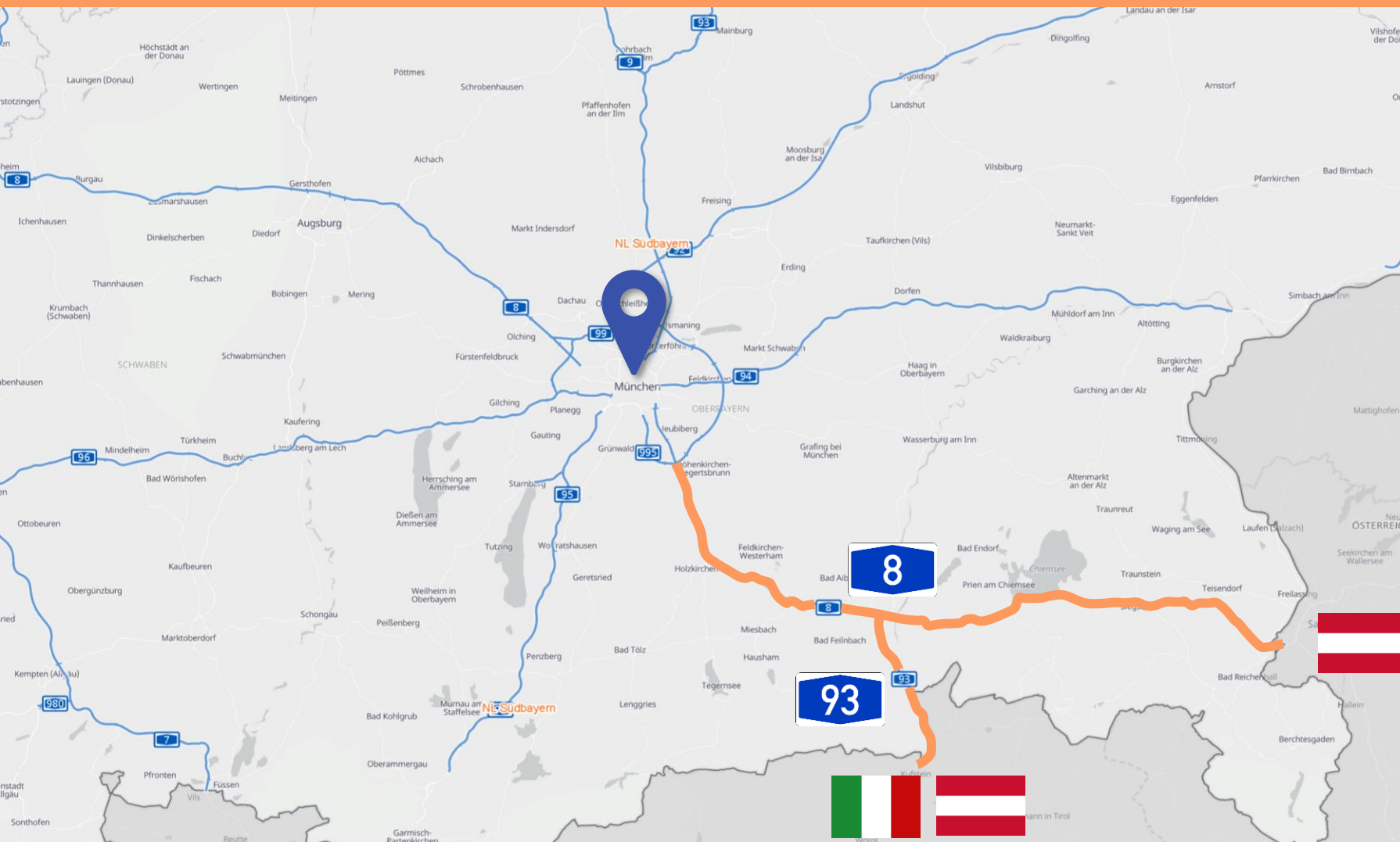




Die Autobahn GmbH des Bundes



Background & User Context



A 8 East and A 93 South

- The A 8 is one of the most important east-west connections in Central Europe
 - Both roads are important connections to Austria, Italy, Croatia and other south-east European countries
 - Crucial routes for holiday and leisure traffic
 - Traffic volumes vary significantly depending on holidays, weather or special events
 - Traffic demand frequently exceeds capacity especially during the summer months
 - An additional challenge lies in necessary capacity-reducing maintenance work in the alpine road network (German as well as Austrian Autobahn)
- This makes trip planning for individual user groups very difficult

Background & User Context

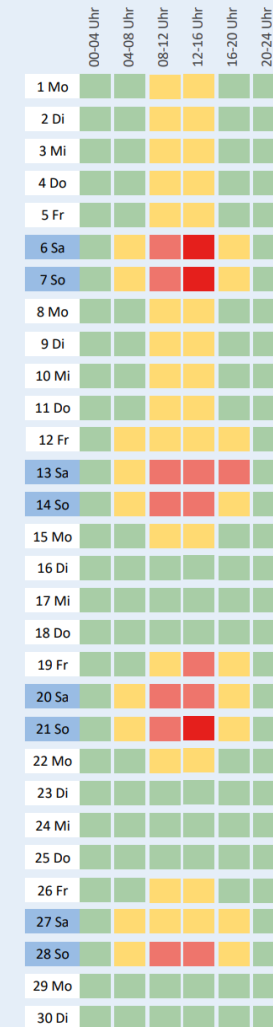
- To support road users, Die Autobahn GmbH has developed a Traffic Calendar (“Fahrkalender”)
- Using historical traffic data
- It provides a traffic forecast for up to two months
- Giving color coded recommendations for trip planning for each day in six time slots
- The goal is to encourage the individual user to shift their planned trips to times with lower traffic demand
- Potential user groups are:
 - Private Traveler / Tourist
 - Local Resident
 - Logistics / Freight / Bus Company
 - Tourism Business
 - Traffic Management Authority

FAHRKALENDER A8 RICHTUNG ÖSTERREICH

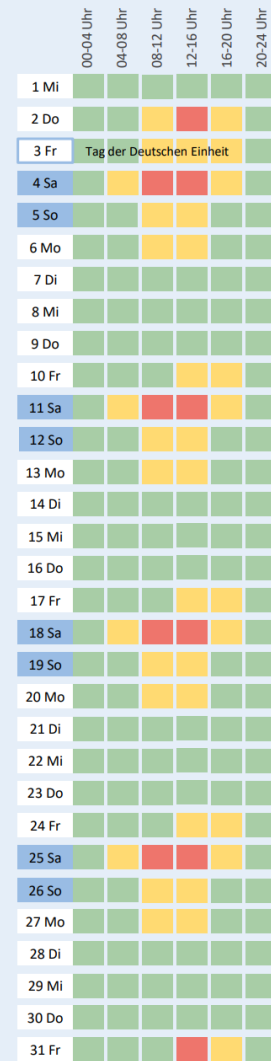
Rosenheim, München, Deutschland → Salzburg, Tauern, Österreich



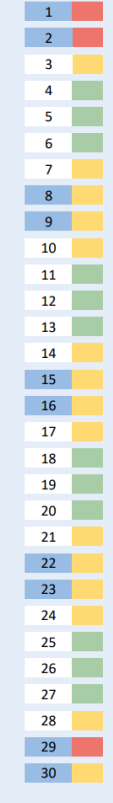
SEPTEMBER



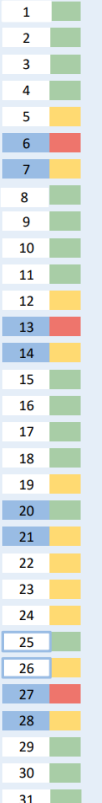
OKTOBER



NOVEMBER



DEZEMBER



■ Wochenende ■ Feiertag
 ■ Flüssiger Verkehr ■ Verstärkter Verkehr
 ■ Starker Verkehr ■ Sehr starker Verkehr

* Die Verkehrsprognosen wurden auf Grundlage eines Verkehrsaufkommens erarbeitet, das durch keine Störungen wie Unfälle, Baustellen, Wetterbedingungen usw. beeinflusst wurde. Als Grundlage für die Einschätzung der erwarteten Verkehrslage wurde unter anderem die Verkehrsnachfrage aus der Vergangenheit in den Prognosehorizont eingeschätzt. Ungeplante spontane Ereignisse und Events wie Unfälle, kurzfristig notwendige Baustellen und witterungsbedingte Einflüsse können die Verkehrslage beeinflussen und sind in der Prognose nicht enthalten.

Challenge Objective

How can historical traffic data and publicly available information be used to automatically generate daily, user-relevant traffic forecasts that support early and informed planning for all these user groups?

Design and prototype a system that:

- Automatically generates daily traffic forecasts
- For all days of 2026–2029
- For the motorway corridors A 8 East and A 93 South (both directions)
- With a focus on holiday and leisure traffic

With the goal to:

- Make holiday travel more predictable and less stressful
- Support better planning for businesses and residents
- Strengthen data-driven public services
- Scale expert knowledge into a sustainable forecasting tool

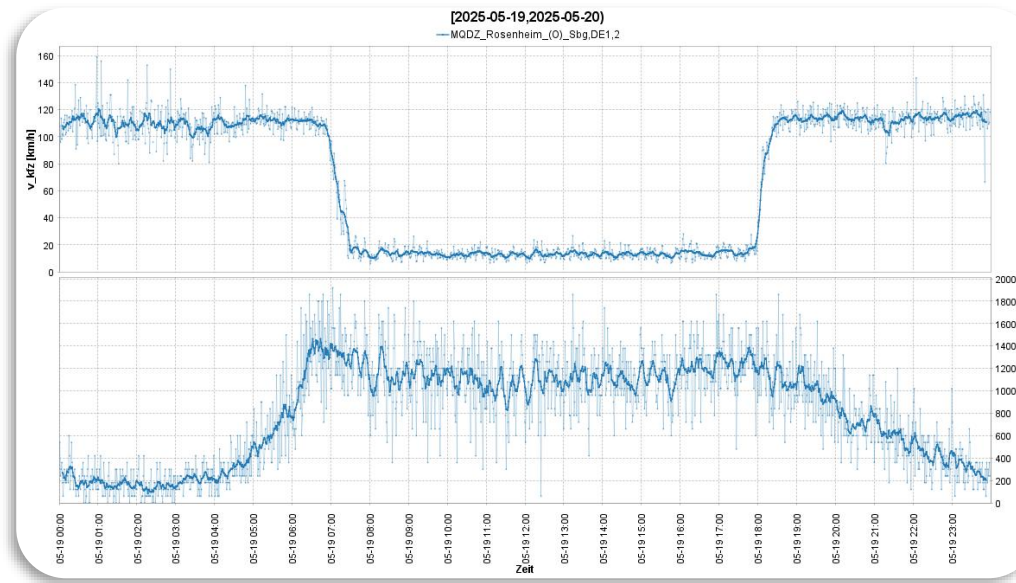
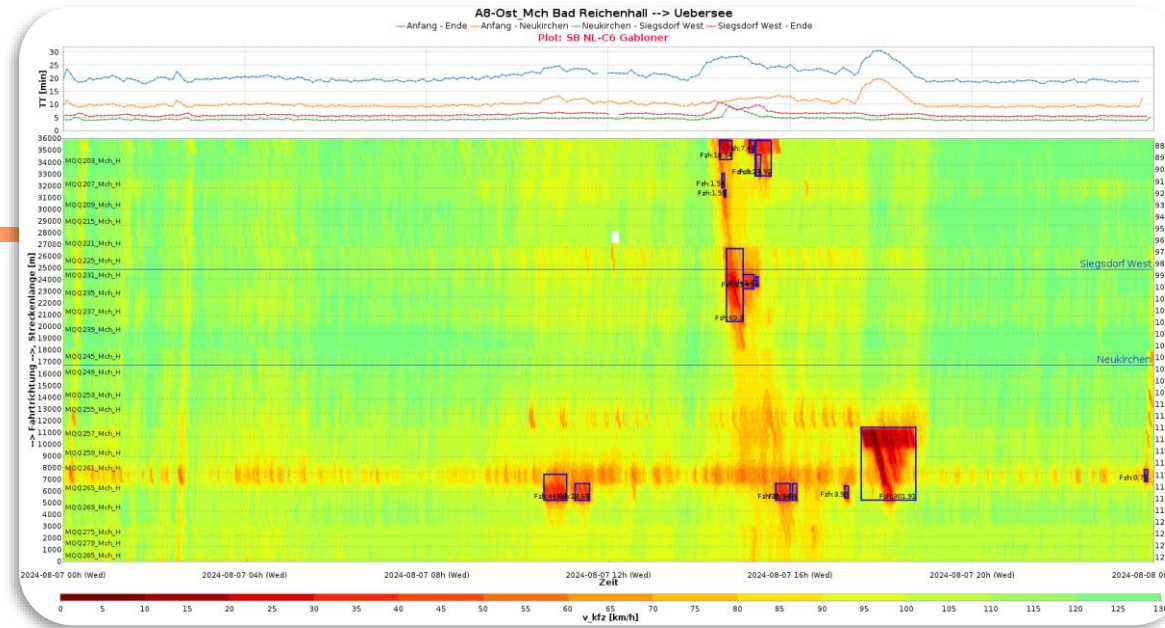
Available Data

Provided by Die Autobahn GmbH:

- Historical traffic data for A 8 and A 93

Publicly Available Data (Optional but Encouraged)

- Weather data
- School holiday calendars (Germany + neighboring countries)
- Public holidays
- Known special events (e.g. border traffic control / ASFiNAG “Dosierung” days)



	A	B	C
1	t_start	q_kfz_h	v_kfz
2	23.03.2026 00:00	180	110.5
3	23.03.2026 00:01	300	108.0
4	23.03.2026 00:02	540	134.9
5	23.03.2026 00:03	180	136.8
6	23.03.2026 00:04	120	106.0
7	23.03.2026 00:05	60	87.0
8	23.03.2026 00:06	0	null
9	23.03.2026 00:07	60	91.0
10	23.03.2026 00:08	120	94.0
11	23.03.2026 00:09	60	122.0
12	23.03.2026 00:10	240	119.0
13	23.03.2026 00:11	360	113.9
14	23.03.2026 00:12	300	119.0
15	23.03.2026 00:13	0	null
16	23.03.2026 00:14	120	141.0
17	23.03.2026 00:15	180	137.0
18	23.03.2026 00:16	0	null
19	23.03.2026 00:17	60	123.0
20	23.03.2026 00:18	60	87.0
21	23.03.2026 00:19	300	132.0
22	23.03.2026 00:20	120	139.0
23	23.03.2026 00:21	120	109.0
24	23.03.2026 00:22	60	87.0

Good luck!

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Quelle: Autobahn GmbH | A81 Neckartalbrücke Weitingen